

DD: SIP-01 — Package Management

Developer implementation guide: DD_SIP-01-Package_Management-DEV_IMPL_GUIDE-v1.0.md.

1. Purpose

This rewritten DD is the developer-facing version of the module contract DD_SIP-01-Package_Management-v1.0.md.

Verdict: ■ Modify (Package-like entities exist in TZ — Offre in Java billing modules + Offer in Laravel onboarding — but need consolidation, geo-scoping, versioning, formal lifecycle) **Wave:** 4 | **Section B:** Subscription / Provisioning

The goal of this rewrite is to make the DD easier for a mid-level developer to implement without losing the detailed source contract.

2. Implementation Scope

This DD should be implemented as part of the owning SOPHIX capability, not as an isolated service unless the deployment architecture explicitly separates that capability.

Implement this DD with these rules:

- Use the owning module APIs/repositories for authoritative writes.
- Use approved internal APIs/client wrappers when reading another module's data.
- Do not query another module's database tables directly from business flow code.
- Treat worker names as logical Camunda external-task topics, not separate deployments.
- One worker application may handle multiple topics, but metrics, retries, and logs must remain visible per topic.
- Emit success events only after the authoritative state commit succeeds.
- Use outbox publishing for Kafka events.

3. Runtime Metadata

Item	Value
Source file	/Users/macbook/Downloads/Sophix_V3_BSS_DDs_MD_2026-05-27/09_catalogs/DD_SIP-01-Package_Management-v1.0.md
Version	v1.0
DD type	module contract
BPMN/process keys	Not explicitly defined
Drools packages	Not explicitly defined

4. Runtime Contracts To Implement

4.1 APIs

- DELETE /api/packages/{id}
- GET /api/packages/{id}
- GET /api/packages/{id}/versions
- GET /api/packages/{id}/versions/{versionId}
- GET /api/packages?status=&franchise=&billingFrequencyDays=&page=&size=
- PATCH /api/packages/:id
- PATCH /api/packages/{id}
- POST /api/packages
- POST /api/packages/:id/versions
- POST /api/packages/{id}/versions

4.2 Tables

- No CREATE TABLE block is explicitly declared in this DD.

For each table in this DD, implement the schema with:

- a short table comment explaining what the table is for;
- column comments or migration notes explaining each field;
- constraints and indexes from the DD;
- seed data where the DD provides operator-specific sample rows.

4.3 Worker Topics

- plm-rules
- plm-session
- subscription-engine

Worker-topic guidance:

- A BPMN service task uses the topic.
- A worker application subscribes to the topic.
- The topic handler validates input variables, calls the owning module/API, writes outputs back to Camunda, and records metrics.
- Do not treat the topic string as a shell command or Kubernetes deployment name.

4.4 Events

- No domain events are explicitly declared in this DD.

Event guidance:

- Write events through the transactional outbox.
- Include operation id, aggregate id, operator code, actor, and correlation data.
- Consumers should be able to rebuild downstream state without querying workflow internals.

5. Developer Implementation Checklist

1. Add or verify database migrations and seed rows.
2. Add or verify config rows for the operator/module.
3. Implement request/command DTOs and validation.
4. Implement idempotency and in-flight operation checks where the DD requires them.
5. Implement Drools fact mapping and package deployment where rules are used.
6. Implement BPMN process, service tasks, gateways, timers, and message catches where the DD is a flow.
7. Implement worker-topic handlers using owning module APIs.
8. Implement compensation paths and manual recovery paths.
9. Emit success/rejected events through the outbox.
10. Add smoke tests for happy path, validation failure, dependency failure, and retry/idempotency.

6. Infrastructure And AWS Notes

Deploy this DD with the existing SOPHIX platform components unless the owning module requires isolation:

Concern	Recommendation
API/runtime	Run inside the owning module deployment or existing workflow API pods.
Workers	Consolidate low-volume topics into shared worker apps; keep per-topic metrics.

Concern	Recommendation
Database	Use the owning module PostgreSQL schema and migration pipeline.
Kafka	Use the foundation Kafka/MSK setup and transactional outbox.
Camunda	Deploy BPMN files through the workflow deployment pipeline.
Drools	Deploy KJAR/rule artifacts through the approved rules pipeline.
Secrets	Use the platform secret manager; on AWS prefer Secrets Manager/Parameter Store with IRSA.
Observability	Alert on stuck operations, failed workers, outbox backlog, and external dependency failures.

7. Original DD Detail Preserved

The following section preserves the detailed source contract for implementation and review.

Verdict: ■ Modify (Package-like entities exist in TZ — Offer in Java billing modules + Offer in Laravel onboarding — but need consolidation, geo-scoping, versioning, formal lifecycle)

Wave: 4 | **Section B:** Subscription / Provisioning

A — Target

1. What this process is about

A **Package** is what customers buy. It groups one or more **Services** (from PLM-CFG-01) at a price, for a billing frequency, available in specific franchises and tech regions. Examples: "GPON Internet 100MB Residential" packages together one Service (the Internet 100MB Service from the catalog) at 5,000 KES/month, available in Nairobi and Mombasa via GPON.

This DD covers the **configuration** of Packages by an admin user: create, read, update, soft-delete, plus audit. There is no workflow here — this is administrative CRUD on catalog data.

Important boundary: SIP-01 owns the Package's **definition** (code, composition, price, geo-scope, wallet/tax bindings). The Package **lifecycle** (Draft → Active → Inactive → EndOfLife, per franchise) is owned by **SIP-02** (Package launch). SIP-01 creates Packages in DRAFT state; transitioning them requires SIP-02 endpoints. This DD focuses on definition only.

PLM scope. PLM owns the catalog of products: Services (PLM-CFG-01), Packages (this DD), Bundles (PLM-CFG-08). PLM-CFG-08 (Bundle) composes Packages from this DD. PLM never calls back into customer/billing modules.

Downstream modules (subscription-engine, fulfillment, billing, search index, etc.) consume Package data via Kafka events and keep their own caches in sync.

2. Business rules

ID	Rule	Triggered on
R-SIP-01-1	Package code must be unique within the catalog (shared namespace with Service and Bundle in PLM-CFG-01 and PLM-CFG-08).	Create, Update
R-SIP-01-2	Package code is immutable once the Package leaves DRAFT state. Code is referenced by external systems (billing records, subscription history, partner integrations) and must be stable.	Update
R-SIP-01-3	A Package must compose at least 1 Service .	Create, Update
R-SIP-01-4	All composed Services must be Active (not retired) at the time the Package is created or updated.	Create, Update
R-SIP-01-5	Package <code>billingFrequencyDays</code> is a positive integer (days per cycle). Typical values: 30 (monthly), 28 (lunar cycle), 15 (prepaid bi-weekly), 7 (weekly), 180 (semi-annual), 365 (annual). The same Package can be cloned for different frequencies; they are separate Package records.	Create, Update
R-SIP-01-6	The Package's <code>targetTechRegions</code> must intersect with the tech regions of each composed Service's <code>equipmentRequirement</code> (an Internet Service requiring GPON ONT can only be packaged for GPON tech regions).	Create, Update

ID	Rule	Triggered on
R-SIP-01-7	Wallet binding precedence: a charge on this Package routes to (1) the Service-level wallet binding if specified, (2) the Package's defaultWalletRef, (3) the system default (TVD for flat-fee, Phone for usage-based). The Package must therefore have a defaultWalletRef consistent with the settlement type of its composed Services.	Create, Update
R-SIP-01-8	A Package cannot be soft-deleted while it is referenced by an active customer subscription.	Delete
R-SIP-01-9	A Package cannot be soft-deleted while it is composed into an active Bundle (coordinated with PLM-CFG-08 rule R-PLM-CFG-08-7).	Delete
R-SIP-01-10	Once a Package has reached ACTIVE state at least once (per SIP-02), composition changes (services[]) are no longer allowed in SIP-01; only pricing or targetFranchises/targetTechRegions adjustments via versioning. Composition changes require retiring the Package and creating a new one.	Update

3. Module owner and impacted modules

Role	Module	What it does in this process
Owner	plm	Owns Package + PackageComposition + PackageVersion data, REST APIs, the package admin UI, the DRL rules KJAR
Impacted (downstream consumers)	subscription-engine (subscription creation references packages), fulfillment (activation reads composition), billing (cycle pricing reads package version), cvm (campaign targeting), search index, BSS Backoffice (catalog views)	Consume PackageCreated/Updated/Retired/VersionPublished Kafka events; refresh their local caches
Coordination	plm (PLM-CFG-08 Bundle)	R-SIP-01-9 — Package delete blocked if referenced by active Bundle; this rule is implemented in the bundle-catalog rule set (same KJAR, same KIE session)
Gateways	—	None used in this process
Engines	Drools KIE	Rule evaluation on every create/update/delete

No workflow, no Camunda. No external system calls. SIP-01 is pure catalog admin.

4. Foundation references

DB → FOUNDATION_DB.md · Drools → FOUNDATION_DROOLS.md · Kafka → FOUNDATION_KAFKA.md · Cache → FOUNDATION_CACHE.md · Auth → FOUNDATION_AUTH.md.

Technical design

Module placement

Layer	Where
Frontend	Shared BSS Backoffice UI (Catalog section, Packages sub-area)
Backend	Existing plm module , sub-area package-catalog/
Rules	Rules Engine (Drools) — plm-rules KJAR (shared with PLM-CFG-01 and PLM-CFG-08; SIP-01 adds its own DRL files and facts)
Events	Internal Event Bus (Kafka)

Single-tenant: same codebase per country, config + rule packages loaded at startup.

plm module — ecosystem (SIP-01 view)

flowchart LR
 UI[BSS Backoffice UI
Catalog section] -->|REST| PLM[plm module
package-catalog]

```

PLM -->|invoke| RE[Rules Engine]
PLM -->|publish| EB[Kafka]
EB --> SUB[subscription-engine cache]
EB --> FUL[fulfillment cache]
EB --> BIL[billing cache]
EB --> CVM[cvm cache]
EB --> SEARCH[search index]
PLM -.FK.-> SVC[Service catalog<br/>PLM-CFG-01<br/>same plm DB]
PLM -.coordinates with.-> BND[Bundle catalog<br/>PLM-CFG-08<br/>same plm DB]

```

plm never calls customer services, billing engine, or fulfillment workflows. Package catalog is purely admin-facing.

Codebase

SIP-01 adds to the existing plm module (created by PLM-CFG-01). Stack stays as chosen for plm.

Java/Spring Boot (continuation of PLM-CFG-01)

```

modules/plm/                                ← existing module
src/main/java/com/sophix/plm/
  service-catalog/                          ← PLM-CFG-01
  bundle-catalog/                          ← PLM-CFG-08
  package-catalog/                         ← SIP-01 (this DD)
    {domain, repository, service, web/rest, event, exception}/
src/main/resources/
  config/liquibase/changelog/plm/
    006_package.xml                        ← new for SIP-01
    007_package_composition.xml           ← new for SIP-01
    008_package_version.xml               ← new for SIP-01

modules/plm-rules/                          ← existing KJAR project
src/main/resources/
  rules/
    sip-01-package-validation.drl         ← R-SIP-01-1, -3, -4, -5, -6, -7, -8, -10
    sip-01-package-immutability.drl       ← R-SIP-01-2
src/main/java/com/sophix/plm/facts/
  PackageFact.java                        ← new
  PackageCompositionFact.java             ← new
  PackageVersionFact.java                 ← new
  ReferencingActiveSubscriptionsFact.java ← new
  ReferencingActiveBundlesFact.java       ← new (coordination with PLM-CFG-08 R-7)

```

Laravel (if stack chosen for plm)

```

modules/plm/
  app/
    Models/
      Package.php                         ← new
      PackageComposition.php              ← new
      PackageVersion.php                  ← new
    Http/Controllers/
      PackageController.php               ← new
    Services/
      PackageCatalogService.php
      PackageValidationService.php        ← calls KIE Server
    Events/
      PackageCreated.php / PackageUpdated.php / PackageRetired.php / PackageVersionPublished.php
  database/migrations/
    2026_05_18_000006_create_package_table.php
    2026_05_18_000007_create_package_composition_table.php
    2026_05_18_000008_create_package_version_table.php

```

Foundation-specific configurations

DB (FOUNDATION_DB.md):

- Cluster: **A** (Config & Catalog) — same as PLM-CFG-01 and PLM-CFG-08
- Database: plm — shares with PLM-CFG-01 and PLM-CFG-08
- New tables (in public schema): package, package_composition, package_version
- New events on audit_log_entry (existing table)
- Messaging tables (messaging.outbox_event, messaging.inbox_event) — already created by PLM-CFG-01, shared

Kafka (FOUNDATION_KAFKA.md):

- Topics published:

- `sophix.plm.package.created`
- `sophix.plm.package.updated`
- `sophix.plm.package.retired`
- `sophix.plm.package.version-published`
- Topics consumed: none
- Key format: `plm:package:{packageId}` (e.g., `plm:package:pkg_01HX...`)
- eventId format: `plm:package:{packageId}:evt_{ulid}`

Cache (FOUNDATION_CACHE.md):

- `plm` does not cache its own data
- Downstream caches use key prefix `sophix:plm:package:` with TTL 24h
- Versions cached separately under `sophix:plm:package-version:{packageId}:{versionId}` (long-lived since versions are immutable once published)
- Sync via Kafka topics above

Drools (FOUNDATION_DROOLS.md):

- KJAR: `plm-rules-{country}_{version}` (shared with PLM-CFG-01, PLM-CFG-08)
- KIE container ID: `plm-rules-ke_1.0.0`
- KIE session: `plm-session`
- New DRL files: `sip-01-package-validation.drl`, `sip-01-package-immutability.drl`
- New facts: `PackageFact`, `PackageCompositionFact`, `PackageVersionFact`, `ReferencingActiveSubscriptionsFact`, `ReferencingActiveBundlesFact`
- Build and deploy: per the reference DD (PLM-CFG-01 — "Building and deploying the rules KJAR")

Camunda: Not used (CRUD only, no workflow).

Frontend

Item	Detail
App	Shared BSS Backoffice UI (this DD adds Packages to the Catalog section, alongside Services and Bundles)
Actor	Product Manager (full CRUD), Billing Lead (full CRUD), Admin (full CRUD), any authenticated role (read-only)
Auth	Bearer JWT (see FOUNDATION_AUTH.md)
Routes	<code>/catalog/packages</code> , <code>/catalog/packages/new</code> , <code>/catalog/packages/:id</code> , <code>/catalog/packages/:id/versions</code> , <code>/catalog/packages/:id/audit</code>

Screens

Package list (/catalog/packages)

- Table: code, name, status (Draft/Active/Inactive/EndOfLife — read-only display; lifecycle controlled by SIP-02), current price, billing frequency, services (count), target franchises, updatedAt, updatedBy
- Filters: status, billing frequency, franchise, search (substring on code/name)
- Pagination, default 20
- Actions: New Package button, row click → edit

Package create / edit (/catalog/packages/new, /catalog/packages/:id)

- Form fields:
- Code (text, required, unique — R-1; immutable once Package leaves DRAFT — R-2)
- Name + display name (text)
- Description (text)

- Services composition (multi-select from PLM-CFG-01 Active Services, min 1 — R-3, R-4)
- Price (number, required) + currency (from country deployment config)
- Billing frequency in days (numeric input; common presets shown as quick buttons: 30 / 28 / 15 / 7 / 180 / 365 — R-5)
- defaultWalletRef (dropdown — R-7)
- defaultTaxGroupRef (dropdown — optional override; falls back to Service-level)
- targetFranchises (multi-select)
- targetTechRegions (multi-select — must intersect with composed Services' tech regions — R-6)
- After Package leaves DRAFT (managed by SIP-02), the form locks code and services fields; only pricing and geo-scope can change (and those changes create a new PackageVersion — see below)
- On Submit: POST /api/packages (new in DRAFT) or PATCH /api/packages/:id (edit)

Package versions (/catalog/packages/:id/versions)

- Read-only timeline of PackageVersion records (effectiveFrom date, price, status: Pending / Active / Superseded)
- Active customers stay on their subscribed version until they upgrade (FUL-04)
- New subscriptions get the latest Active version
- Admin can publish a new version: POST /api/packages/:id/versions (changes price / geo-scope, not composition; creates a new PackageVersion record)

Package audit (/catalog/packages/:id/audit)

- Timeline of audit_log_entry rows for this Package
- Each row: timestamp, actor, action (CREATE/UPDATE/RETIRE/VERSION_PUBLISHED), expandable diff
- Read-only

Client-side validation

Basic input only (required fields, format). Server is authoritative for all business rules (R-1 through R-10). The dev should NOT replicate Drools logic client-side.

Data model

```
classDiagram
    class Package {
        PackageId id
        String code
        String name
        String displayName
        String description
        PackageStatus status
        Integer billingFrequencyDays
        String defaultWalletRef
        String defaultTaxGroupRef
        List-String~ targetFranchises
        List-String~ targetTechRegions
        PackageVersionId currentVersionId
    }
    class PackageComposition {
        PackageId packageId
        ServiceId serviceId
        Int sequence
    }
    class PackageVersion {
        PackageVersionId id
        PackageId packageId
        Money price
        Date effectiveFrom
        VersionStatus status
    }
    class Service {
        ServiceId id
        String code
        ...
    }
    Package "1" --> "n" PackageComposition
    PackageComposition --> Service
    Package "1" --> "n" PackageVersion
```

Package --> PackageVersion : currentVersionId

Tables (PostgreSQL, database plm, schema public)

package — Package catalog

Column	Type	Notes			
id	TEXT PK	ULID, e.g. pkg_01HX...			
code	TEXT NOT NULL	unique across plm catalog (R-1); immutable post-DRAFT (R-2)			
name	TEXT NOT NULL	internal name, e.g. "Internet 100MB Residential"			
display_name	TEXT	customer-facing display name			
description	TEXT				
status	TEXT NOT NULL DEFAULT 'DRAFT'	DRAFT \	ACTIVE \	INACTIVE \	END_OF_LIFE (per franchise: see SIP-02; this column is the master status across all franchises — current minimum)
billing_frequency_ days	INT NOT NULL	positive integer; days per billing cycle (R-5). Typical: 30, 28, 15, 7, 180, 365.			
default_wallet_ref	TEXT NOT NULL	string ref (R-7)			
default_tax_group_ ref	TEXT NULL	string ref; falls back to Service-level if NULL			
target_franchises	TEXT[] NOT NULL DEFAULT '{}'	franchise codes			
target_tech_region s	TEXT[] NOT NULL DEFAULT '{}'	tech region codes			
current_version_id	TEXT NULL	FK → package_versi on.id; the active version for new subscriptions			
created_at	TIMESTAMP NOT NULL DEFAULT NOW()				
created_by	TEXT NOT NULL				
updated_at	TIMESTAMP				
updated_by	TEXT				
retired_at	TIMESTAMP NULL	soft delete			

Indexes:

- (code) WHERE retired_at IS NULL UNIQUE
- (status) for status-filtered queries
- GIN index on (target_franchises) for franchise-availability lookups

package_composition — Package → Services many-to-many

Column	Type	Notes
package_id	TEXT NOT NULL	FK → package.id ON DELETE CASCADE
service_id	TEXT NOT NULL	FK → service.id (in same plm DB; PLM-CFG-01)
sequence	INT NOT NULL DEFAULT 0	display order

Primary key: (package_id, service_id). Indexes:

- (service_id) for reverse lookups (PLM-CFG-01 R-5: is this Service referenced by any active Package?)

package_version — Pricing/geo-scope versions of a Package

Column	Type	Notes		
id	TEXT PK	ULID, e.g. pkv_01HX...		
package_id	TEXT NOT NULL	FK → package.id		
price	NUMERIC(12,2) NOT NULL	price at this version		
currency	TEXT NOT NULL	ISO 4217 (e.g., KES)		
target_franchises	TEXT[] NOT NULL DEFAULT '{}'	snapshot of franchises for this version		
target_tech_regions	TEXT[] NOT NULL DEFAULT '{}'	snapshot of tech regions for this version		
effective_from	TIMESTAMP NOT NULL	when this version takes effect for new subscriptions		
effective_until	TIMESTAMP NULL	when this version was superseded (set when next version becomes active)		
status	TEXT NOT NULL	PENDING \	ACTIVE \	SUPERSEDED
created_at	TIMESTAMP NOT NULL DEFAULT NOW()			
created_by	TEXT NOT NULL			

Indexes:

- (package_id, effective_from DESC) for version timeline queries
- (package_id, status) for finding the current active version

Versioning model: Package definition (code, name, composition, wallet/tax bindings) lives on the package row. **Pricing and geo-scope** changes create a new package_version row. The package.current_version_id points at the active version. When the admin publishes a new version, the previous one transitions to SUPERSEDED and the new one becomes ACTIVE at its effective_from. Active customer subscriptions reference a specific package_version_id and remain on that version until they upgrade (FUL-04) — preserving historical pricing.

Rules (Drools)

Rule IDs match the business-rule table in Section 2. Full DRL implementation below.

sip-01-package-validation.drl

```
package com.sophix.plm.rules

import com.sophix.plm.facts.PackageFact
import com.sophix.plm.facts.PackageCompositionFact
import com.sophix.plm.facts.ExistingCodesFact
import com.sophix.plm.facts.ReferencingActiveSubscriptionsFact
import com.sophix.plm.facts.ReferencingActiveBundlesFact
import com.sophix.plm.facts.ValidationError

// R-SIP-01-1 – Package code must be unique across plm catalog

rule "R-SIP-01-1-code-uniqueness"
when
    $p: PackageFact($code: code, $id: id, action != "DELETE")
    $existing: ExistingCodesFact(codes contains $code, ownerId != $id)
then
    insert(new ValidationError("R-SIP-01-1",
                               "code",
                               "Code already exists in catalog: " + $code));
end

// R-SIP-01-3 – Package must compose at least 1 Service

rule "R-SIP-01-3-minimum-composition"
when
    $p: PackageFact($id: id, action != "DELETE")
    $c: PackageCompositionFact(packageId == $id, services.size() < 1)
then
    insert(new ValidationError("R-SIP-01-3",
                               "services",
                               "Package must compose at least 1 Service"));
end

// R-SIP-01-4 – All composed Services must be Active
```

```

rule "R-SIP-01-4-services-must-be-active"
when
    $p: PackageFact($id: id, action != "DELETE")
    $c: PackageCompositionFact(packageId == $id, retiredServices.size() > 0)
then
    insert(new ValidationError("R-SIP-01-4",
                              "services",
                              "Composed Services are retired: " + $c.getRetiredServices()));
end

// R-SIP-01-5 – billingFrequencyDays must be a positive integer

rule "R-SIP-01-5-billing-frequency-positive"
when
    $p: PackageFact($freq: billingFrequencyDays, action != "DELETE")
    eval($freq == null || $freq <= 0)
then
    insert(new ValidationError("R-SIP-01-5",
                              "billingFrequencyDays",
                              "billingFrequencyDays must be a positive integer; got " + $freq));
end

// R-SIP-01-6 – targetTechRegions must intersect composed Services' tech regions

rule "R-SIP-01-6-tech-region-intersection"
when
    $p: PackageFact($id: id, action != "DELETE")
    $c: PackageCompositionFact(packageId == $id, techRegionMismatch == true)
then
    insert(new ValidationError("R-SIP-01-6",
                              "targetTechRegions",
                              "Package targetTechRegions must intersect with each composed Service's tech regions"));
end

// R-SIP-01-7 – defaultWalletRef must be consistent with composed Services' settlement model

rule "R-SIP-01-7-wallet-consistency"
when
    $p: PackageFact($id: id, action != "DELETE")
    $c: PackageCompositionFact(packageId == $id, walletInconsistency == true)
then
    insert(new ValidationError("R-SIP-01-7",
                              "defaultWalletRef",
                              "defaultWalletRef inconsistent with composed Services' settlement model"));
end

// R-SIP-01-8 – Soft-delete blocked if active subscriptions reference this Package

rule "R-SIP-01-8-delete-blocked-by-subscription"
when
    $p: PackageFact($id: id, action == "DELETE")
    $r: ReferencingActiveSubscriptionsFact(packageId == $id, activeSubscriptions.size() > 0)
then
    insert(new ValidationError("R-SIP-01-8",
                              "id",
                              "Package is referenced by " + $r.getActiveSubscriptions().size() + " active subscriptions"));
end

// R-SIP-01-9 – Soft-delete blocked if Package is composed into an active Bundle
// Coordinated with PLM-CFG-08 R-7

rule "R-SIP-01-9-delete-blocked-by-bundle"
when
    $p: PackageFact($id: id, action == "DELETE")
    $r: ReferencingActiveBundlesFact(packageId == $id, activeBundles.size() > 0)
then
    insert(new ValidationError("R-SIP-01-9",
                              "id",
                              "Package is composed into active Bundles: " + $r.getActiveBundles()));
end

```

sip-01-package-immutability.drl

```

package com.sophix.plm.rules

import com.sophix.plm.facts.PackageFact
import com.sophix.plm.facts.ValidationError

// R-SIP-01-2 – Code is immutable once Package leaves DRAFT

rule "R-SIP-01-2-code-immutable-after-draft"
when
    $p: PackageFact(action == "UPDATE", $newCode: code, $oldCode: previousCode, $oldStatus: previousStatus)

```

```

        eval(!$newCode.equals($oldCode) && !"DRAFT".equals($oldStatus))
    then
        insert(new ValidationError("R-SIP-01-2",
                                   "code",
                                   "Package code cannot be changed after Package has left DRAFT state"));
    end

    // R-SIP-01-10 – Composition is immutable once Package has been ACTIVE at least once

    rule "R-SIP-01-10-composition-immutable-post-active"
    when
        $p: PackageFact(action == "UPDATE", $hasBeenActive: hasBeenActive == true, $compositionChanged: compositionChanged ==
    then
        insert(new ValidationError("R-SIP-01-10",
                                   "services",
                                   "Package composition cannot be changed once Package has been ACTIVE. Retire and create a ne
    end

```

Fact classes (POJOs in com.sophix.plm.facts)

Convention: Lombok @Data + @NoArgsConstructor + @AllArgsConstructor as established in PLM-CFG-01.

PackageFact.java

```

package com.sophix.plm.facts;

import lombok.AllArgsConstructor;
import lombok.Data;
import lombok.NoArgsConstructor;

import java.util.List;

@Data
@NoArgsConstructor
@AllArgsConstructor
public class PackageFact {
    private String id;
    private String code;
    private String previousCode;           // for R-2 (compare new vs old)
    private String name;
    private String status;
    private String previousStatus;         // for R-2 (was it DRAFT before?)
    private boolean hasBeenActive;         // for R-10
    private boolean compositionChanged;    // for R-10
    private Integer billingFrequencyDays;
    private String defaultWalletRef;
    private String defaultTaxGroupRef;
    private List<String> targetFranchises;
    private List<String> targetTechRegions;
    private String action; // "CREATE" | "UPDATE" | "RETIRE" | "DELETE"
}

```

PackageCompositionFact.java

```

package com.sophix.plm.facts;

import lombok.AllArgsConstructor;
import lombok.Data;
import lombok.NoArgsConstructor;

import java.util.List;

@Data
@NoArgsConstructor
@AllArgsConstructor
public class PackageCompositionFact {
    private String packageId;
    private List<String> services;           // serviceIds composed into this Package
    private List<String> retiredServices;    // for R-4
    private boolean techRegionMismatch;     // for R-6
    private boolean walletInconsistency;    // for R-7
}

```

PackageVersionFact.java (used by version publish flow — minor; full validation lives in SIP-02 for lifecycle, this is for SIP-01 version-publish only)

```

package com.sophix.plm.facts;

import lombok.AllArgsConstructor;
import lombok.Data;
import lombok.NoArgsConstructor;

import java.math.BigDecimal;

```

```

import java.time.Instant;

@Data
@NoArgsConstructor
@AllArgsConstructor
public class PackageVersionFact {
    private String id;
    private String packageId;
    private BigDecimal price;
    private String currency;
    private Instant effectiveFrom;
    private String status;
}

```

ReferencingActiveSubscriptionsFact.java

```

package com.sophix.plm.facts;

import lombok.AllArgsConstructor;
import lombok.Data;
import lombok.NoArgsConstructor;

import java.util.List;

@Data
@NoArgsConstructor
@AllArgsConstructor
public class ReferencingActiveSubscriptionsFact {
    private String packageId;
    private List<String> activeSubscriptions; // subscription IDs from subscription-engine (string refs)
}

```

ReferencingActiveBundlesFact.java

```

package com.sophix.plm.facts;

import lombok.AllArgsConstructor;
import lombok.Data;
import lombok.NoArgsConstructor;

import java.util.List;

@Data
@NoArgsConstructor
@AllArgsConstructor
public class ReferencingActiveBundlesFact {
    private String packageId;
    private List<String> activeBundles; // bundleIds where this Package is composed (local lookup in same plm DB)
}

```

(ExistingCodesFact and ValidationError were defined in PLM-CFG-01 and are reused.)

The caller's responsibility:

- For **create/update**: insert PackageFact + PackageCompositionFact (with composition state computed by the service layer, including retiredServices, techRegionMismatch, walletInconsistency) + ExistingCodesFact; fire rules; collect ValidationError.
- For **delete**: insert PackageFact (action="DELETE") + ReferencingActiveSubscriptionsFact (looked up from subscription-engine API) + ReferencingActiveBundlesFact (local query in same plm DB); fire rules.
- For **version publish**: insert PackageFact + PackageVersionFact; fire rules.

Build and deploy the KJAR: per the reference DD (PLM-CFG-01 — "Building and deploying the rules KJAR").

Events (Kafka)

Event	Topic	When	Consumers
PackageCreated	sophix.plm.package.created	After successful insert (in DRAFT)	subscription-engine cache, fulfillment cache, billing cache, search index
PackageUpdated	sophix.plm.package.updated	After successful update	All consumers above
PackageRetired	sophix.plm.package.retired	After soft-delete	All consumers above
PackageVersionPublished	sophix.plm.package.version-published	After new version published	billing cache (prices), subscription-engine cache (new-subscription pricing), search index

Key format: plm:package:{packageId} · eventId format: plm:package:{packageId}:evt_{ulid}.

PackageCreated envelope:

```
{
  "eventId": "plm:package:pkg_01HX2Y3Z4:evt_01JZ5K6...",
  "eventType": "PackageCreated",
  "aggregateId": "pkg_01HX2Y3Z4",
  "timestamp": "2026-05-18T10:23:00Z",
  "actor": "alex.oyuga",
  "correlationId": "corr_01HX...",
  "payload": {
    "packageId": "pkg_01HX2Y3Z4",
    "code": "INET_100_RES",
    "name": "Internet 100MB Residential",
    "displayName": "Home Fast 100",
    "status": "DRAFT",
    "billingFrequencyDays": 30,
    "defaultWalletRef": "TVD",
    "defaultTaxGroupRef": "Internet-ExciseAndVAT",
    "services": ["svc_gpon_internet_100mb"],
    "targetFranchises": ["Nairobi", "Mombasa"],
    "targetTechRegions": ["GPON"]
  }
}
```

PackageVersionPublished envelope:

```
{
  "eventId": "plm:package:pkg_01HX2Y3Z4:evt_01JZ5N3...",
  "eventType": "PackageVersionPublished",
  "aggregateId": "pkg_01HX2Y3Z4",
  "timestamp": "2026-05-18T16:10:00Z",
  "actor": "alex.oyuga",
  "correlationId": "corr_01HX...",
  "payload": {
    "packageId": "pkg_01HX2Y3Z4",
    "versionId": "pkv_01JZ5N3...",
    "price": 5500,
    "currency": "KES",
    "effectiveFrom": "2026-06-01T00:00:00Z",
    "supersedesVersionId": "pkv_01HX..."
  }
}
```

PackageUpdated and PackageRetired follow the same patterns as PLM-CFG-01 (changedFields diff for update, packageId + retiredAt for retire).

Cache strategy: plm does not cache its own data. Downstream caches use key prefix sophix:plm:package;; PackageVersion under sophix:plm:package-version:. See FOUNDATION_CACHE.md.

Exceptions

Standard taxonomy (per PLM-CFG-01).

APIs (in plm/package-catalog/)

Auth (per FOUNDATION_AUTH.md):

- From UI: Bearer JWT; role from claims enforces per-endpoint permissions
- From other modules: X-Service-Token header
- Either auth path accepted on read endpoints

Method + Path	Purpose	Auth role
POST /api/packages	Create (in DRAFT)	Product Manager, Admin
GET /api/packages/{id}	Read	any authenticated
GET /api/packages?status=&franchise=&billingFrequencyDays=&page=&size=	List/filter	any authenticated
PATCH /api/packages/{id}	Update	Product Manager, Admin
DELETE /api/packages/{id}	Soft-delete	Product Manager, Admin
GET /api/packages/{id}/versions	List version history	any authenticated
POST /api/packages/{id}/versions	Publish new version (pricing/geo-scope change)	Product Manager, Admin

Method + Path	Purpose	Auth role
GET /api/packages/{id}/versions/{versionId}	Read specific version	any authenticated

Status codes (same as PLM-CFG-01)

200 read/update · 201 create · 204 delete · 400 input validation · 401 unauth · 403 forbidden · 404 not found · 409 business rule conflict · 500 internal · 502/503/504 backend.

POST /api/packages

Request:

```
{
  "code": "INET_100_RES",
  "name": "Internet 100MB Residential",
  "displayName": "Home Fast 100",
  "description": "100Mbps residential GPON internet",
  "billingFrequencyDays": 30,
  "defaultWalletRef": "TVD",
  "defaultTaxGroupRef": "Internet-ExciseAndVAT",
  "serviceIds": ["svc_gpon_internet_100mb"],
  "price": 5500,
  "currency": "KES",
  "targetFranchises": ["Nairobi", "Mombasa"],
  "targetTechRegions": ["GPON"]
}
```

The Package is created in DRAFT with an initial PackageVersion (PENDING until SIP-02 activates the Package).

201 response:

```
{
  "id": "pkg_01HX2Y3Z4",
  "code": "INET_100_RES",
  "status": "DRAFT",
  "currentVersionId": "pkv_01HX2Y3Z5",
  "currentVersion": {
    "id": "pkv_01HX2Y3Z5",
    "price": 5500,
    "currency": "KES",
    "status": "PENDING"
  },
  ...
}
```

409 (R-1 — duplicate code):

```
{
  "error": "BUSINESS_RULE_ERROR",
  "rule": "R-SIP-01-1",
  "message": "Code already exists in catalog: INET_100_RES",
  "details": { "field": "code" }
}
```

400 (R-4 — retired service in composition):

```
{
  "error": "BUSINESS_RULE_ERROR",
  "rule": "R-SIP-01-4",
  "message": "Composed Services are retired",
  "details": { "field": "services", "retiredServices": ["svc_old_gpon"] }
}
```

POST /api/packages/{id}/versions

Used to publish a new pricing/geo-scope version of an existing Package (without changing composition).

Request:

```
{
  "price": 5800,
  "currency": "KES",
  "targetFranchises": ["Nairobi", "Mombasa", "Kisumu"],
  "targetTechRegions": ["GPON"],
  "effectiveFrom": "2026-06-01T00:00:00Z"
}
```

201 response:

```
{
  "id": "pkv_01JZ5N3...",
}
```

```

    "packageId": "pkg_01HX2Y3Z4",
    "price": 5800,
    "currency": "KES",
    "effectiveFrom": "2026-06-01T00:00:00Z",
    "status": "PENDING"
}

```

On effectiveFrom, a scheduled job (or the next request after the cutover) transitions the new version to ACTIVE and the previous to SUPERSEDED. Active subscriptions stay on their stored packageVersionId until they upgrade (FUL-04).

DELETE /api/packages/{id}

Fires R-8 and R-9.

204 on success.

409 (R-8 — active subscriptions reference Package):

```

{
  "error": "BUSINESS_RULE_ERROR",
  "rule": "R-SIP-01-8",
  "message": "Package is referenced by 412 active subscriptions",
  "details": { "activeSubscriptionCount": 412 }
}

```

409 (R-9 — Package composed into active Bundle):

```

{
  "error": "BUSINESS_RULE_ERROR",
  "rule": "R-SIP-01-9",
  "message": "Package is composed into active Bundles",
  "details": { "activeBundles": ["bnd_triple_play_pro"] }
}

```

B — Current TZ

Two parallel Package-like entities exist in TZ codebase:

- Offre (Java) — in sofixAccountModule and duplicated across 2-3 other Java modules. Conflates Service + Package roles. JHipster CRUD via OffreResource. Has price, services, billing frequency.
- Offer (Laravel) — in onboarding module. Used during subscription capture. Stores selected offer ID; queries Java module for details.

Neither has:

- Per-franchise availability (TZ is single deployment)
- Tech region constraints
- Formal Draft/Active/Inactive/EndOfLife states
- Versioning (price changes overwrite, breaking historical subscriptions)
- Bundle coordination
- Kafka events on package changes
- Drools-based validation

Wallet/tax bindings exist via separate BindWalletOffer / BindTaxOffer entities (Java); these need to migrate into the Package entity itself.

```

classDiagram
    class Offre {
        Long id
        String code
        String nom
        Float prix
        String frequenceFacturation
    }
    class BindWalletOffer {
        Long offreId
        Long walletId
    }
    class BindTaxOffer {
        Long offreId
        Long taxId
    }
    class Offer {

```

```

    Long id
    String code
    Long sourceOffreId
}
Offre --> BindWalletOffre
Offre --> BindTaxOffre
Offer -.points to.-> Offre

```

C — Gap & bridge

Aspect	TZ today	Target	Bridge
Package entity	Offre (Java) + Offer (Laravel duplicate)	Single Package entity in plm module	Consolidate; migrate data from Offre → package table; deprecate Offer in onboarding (reference Package by ID instead)
Composition	OffreService join	package_composition join	Reuse pattern; migrate data
Wallet/Tax binding	Separate BindWalletOffre / BindTaxOffre tables	Inline defaultWalletRef / defaultTaxGroupRef on Package	Migrate bindings into Package columns; drop bind tables after migration
Versioning	None (overwrites break history)	package_version table	New table; backfill: create one PackageVersion for each existing Offre at current price
Geo-scope	None	target_franchises[] + target_tech_regions[] arrays	New columns; populate with default (all franchises in current deployment) for existing packages during migration
Lifecycle	Status field, ad-hoc	DRAFT / ACTIVE / INACTIVE / END_OF_LIFE — owned by SIP-02	This DD adds the status column with DRAFT default; SIP-02 owns transitions
Validation rules	Hand-coded in services	Drools KJAR with 10 rules	New DRL files
Events	None	4 Kafka topics	New publishers using outbox pattern
Admin UI	JHipster pages	BSS Backoffice Catalog section	New screens; deprecate JHipster pages after migration
Coordination with Bundle	None	R-9 fires from Package delete	Coordinated with PLM-CFG-08 R-7

Dependencies

- **PLM-CFG-01 (Service Catalog)** — required. Packages compose Services. **Already completed.**
- **PLM-CFG-08 (Bundle Catalog)** — required for R-9 coordination. **Already completed.**
- **PLM-CFG-02 (Tax configuration)** — required for defaultTaxGroupRef validation. Pending.
- **PLM-CFG-03 (Wallet configuration)** — required for defaultWalletRef validation. Pending.
- **SIP-02 (Package launch & lifecycle)** — paired DD; this DD creates Packages in DRAFT; SIP-02 manages transitions. Both can be developed in parallel since SIP-02 only reads from this DD's tables.
- Foundations (DB, Kafka, Drools, Cache, Auth) — all in place.